

Amanah Trader

Autonomous Shariah-Compliant Options Agent

Deterministic gates → LLM proposes, hard rules decide. Live on **PA3W2J1H6I3X** — **\$100,418.59 · +0.42%**, broker-confirmed. No human approval after deploy.

Python + Alpaca CLI

Qwen3-27B proposer only

✓ 83 tests

amanahtrader.uk/hackathon

Prior work github.com/SalmonIsFish/Ai_Finance_Syariah used as spec only — every file authored Aug 28–Sep 4, verified by commit dates.

No interest. No ambiguity. No gambling.

Islamic commercial law screens capital before it screens returns. Wealth has to come from real trade, risk-sharing, or productive effort — never from lending at interest, never from a contract too ambiguous to evaluate, never from a wager on chance. An agent that claims to honor this mandate has to **prove all three**, not describe them.

 RIBA — INTEREST

Profit must come from real economic activity or shared risk, not from lending money at a fixed rate.

Rules out: bonds, margin financing, leveraged borrowing, interest-bearing holdings.

 GHARAR — UNCERTAINTY

What's being bought, its price, and the ability to deliver it must be knowable to both sides at the time of the contract.

Rules out: one-sided quotes, unpriced or undeliverable positions, terms too ambiguous to assess.

 MAYSIR — GAMBLING

Wealth must be earned through effort, trade, or shared enterprise risk — not a zero-sum wager decided by chance.

Rules out: naked speculation, positions with no capacity or intent to own the underlying.

Most autonomous trading agents ignore this mandate entirely. This one enforces all three — in code, on every single trade, not as a stated policy. What that looks like in practice ↓

Halal investing stops at screening. Nobody screens the trade.

 THE MARKET

~1.9B Muslims globally; Islamic finance assets estimated near **\$4 trillion** and still growing double-digit. No US-listed broker offers an actively-managed, per-trade-auditable halal options product — the category is effectively empty.

 THE CUSTOMER

Retail Muslim investors who want **yield**, not just a purity filter on their brokerage app — and Islamic fintechs / robo-advisors who'd license a compliance engine rather than build gate logic from scratch.

 THE GAP

Existing "Shariah-compliant" products — **Wahed Invest, Zoya** — screen **once, at onboarding** — a static list of tickers. None re-verify riba/gharar/maysir per trade, in real time, so none can prove a specific position was permissible — only that a stock was once on a list.

Business model: license this gate engine to Islamic fintechs as a per-trade verification API, or run it directly as a fee-based managed income strategy — not just a demo bot.

10PM MYT is market open. Ethics can't be a prompt.

 PROBLEM

Human-approval = not autonomous. Old bot required typing `EXECUTE PAPER` at 10PM. Judging scores autonomy + robustness.

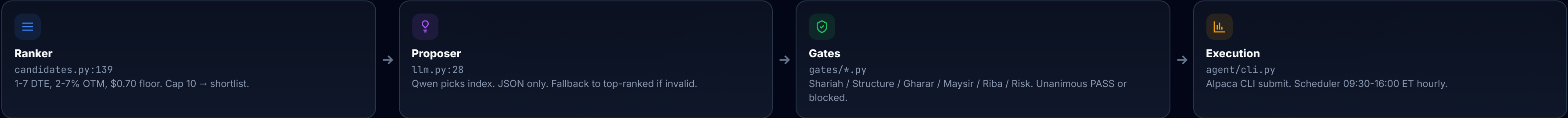
LLM hallucination risk: one invented strike = non-compliant trade with no recall.

 OUR ANSWER

Fail-closed gates. LLM never decides compliance. Six pure functions — Shariah, Structure, **Gharar**, **Maysir**, **Riba**, Risk — cover all three prohibitions of Islamic commercial law; any REJECT blocks submit.

LLM only writes a 1-3 sentence *rationale* (always positive) — the red `Gate block` is the verdict.

Propose, then veto.



Why an LLM at all, if it can't override a gate? A static formula optimizes one axis; the Proposer weighs OTM cushion, premium, and days-to-expiry together, the way a trader reads a chain — judgment that matters more as the shortlist grows past today's 10 candidates, not less.

15 symbols, not 12.

Low-strike = more basket.

UNIVERSE

added Sep 1

AAPL

NVDA

GOOGL

MSFT

CSCO

COST

PG

JNJ

HD

ORCL

ADBE

CRM

INTC

PFE

KO

New adds ~\$26 / ~\$24 / ~\$60 — collateral \$2-6k vs \$21-33k. Tech 8 · Staples 3 · Health 2.

Avg confidence

85.1

EXAMPLE SCREENS

SYMBOL	SIGNAL	RESULT
INTC	82% Semiconductor mfg, debt 0.14	PASS
PFE	84% Pharma, debt 0.20	PASS
KO	86% Staples, debt 0.24	PASS
Ex.	55% REVIEW — below 70% threshold	REJECT

MSCI Islamic methodology, hand-scored. REVIEW 50-69 and FAIL both REJECT — no LLM override. (The 688-record SC-SAC databank committed at `data/malaysian_shariah_databank.json` in the repo lists Bursa Malaysia tickers and cannot classify US symbols; it is retained as data-governance evidence, not consulted by this gate.)

Tighter OTM, higher floor, premium second.

BAND TUNE Sep 1, A+B

target_otm — closer = more premium

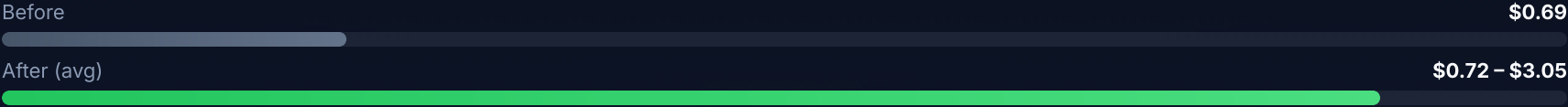
4% → 3%

min_premium — filters \$0.23 CSCO noise

\$0.05 → \$0.70

spread ≤15% · 1-7 DTE · cap 10 · diversity guard (no same underlying same day)

EFFECT ON LIVE PICKS



8 fills since the tune, avg \$1.42 — we pitch **premium collected + MTM**, not chase.

11 fills, 8 refusals, the log tells the truth.

DECISION TIMELINE Sep 3, ET

- 09:35:01 **AUTO-CLOSE AAPL/GOOGL puts** — bought-to-close 3 legs at market open **cron, no human**
- 09:36:40 **SUBMITTED NVDA 222.5P** — 6 DTE, 2.41% OTM, \$1.74 **PASS**
- 10:38:39 **SUBMITTED AAPL 320P** — 6 DTE, 2.89% OTM, \$1.15 **PASS**
- 11:39-12:39 **SUBMITTED INTC 88P×3, 87P×2** — 6 DTE, 2.6-3.9% OTM, \$0.72-\$1.15 **PASS**
- 13:39-15:40 **NO_CANDIDATES ×3** — \$3,006 deployable, nothing fits **capacity-gated**

P&L ATTRIBUTION dashboard · scale \$0-\$1,000

Premium		+\$817
MTM		+\$180
Equity		+\$418.59 (0.42%)

Collateral **\$98,050** locked against **\$101,056** cash — genuinely cash-secured, **\$3,006** options buying power free.
Reported from `position_list`, never from submitted orders: 11 submitted since Monday, 4 currently open — the rest closed automatically at market open via the EOD script, and the gap is published.

Runs while you sleep.

 SCHEDULER

`hackathon-scheduler.service`
09:30-16:00 ET hourly, ZoneInfo ET, sleeps weekends.

No human click since Mon 09:30.

 OBSERVABILITY

`/api/metrics`
gate_pass **40.5%**
rejected_by_gate `{risk:8}`
premium_by_underlying `{AAPL 130, INTC 494, NVDA 193}`
orders_submitted `11` → orders_open `4`
(denominator grows as capacity-limited `NO_CANDIDATES` cycles accumulate — not new rejections;
capacity is now \$3.0k after 4 fills, no remaining candidate clears its collateral floor)
Nightly `logs/report-YYYY-MM-DD.md`.

 AUDIT TRAIL

Now shows `Gate block: RISK: orders_today_cap` red above the always-positive
Proposer pick — mobile-readable without tooltip.

Riba, gharar, maysir. All enforced in code.

THE OBJECTION, STATED AGAINST OURSELVES

Options are named in the *gharar* and *maysir* literature as examples of what is **prohibited**, and **AAOIFI generally prohibits conventional options outright**. We say so in our own docs rather than routing around it.

The objections turn on two words: *naked*, and *pure* speculation. A fully cash-secured put on a screened business, struck below market, is a different act — and each narrowing condition is a gate, not a claim.

`docs/shariah/position-cash-secured-puts.md` — engineering rationale, **not a fatwa**.

ADVERSARIAL EVIDENCE deterministic

- **Richest premium on the board** unlisted symbol **shariah**
- **Naked put** no delivery capacity **maysir**
- **One-sided quote, no ask** price is indeterminate **gharar**
- **Book already 2× levered** real production state **structure**
- **Normal conditions** all six gates **PASS**

`docs/backtest/gate-stress-report.md` — 20 scenarios against the *real* gate functions, no mocks, no network, no clock. **19 refused, 1 allowed**. Byte-identical on every run.

Update Sep 3: capital gating lifted after the EOD auto-close freed collateral — NVDA/AAPL/INTC fills that morning cleared gharar (spread_pct, delivery_funded), maysir (would_take_delivery), and riba (broker_margin_available) with real numbers in `logs/decisions.jsonl`, not just the stress harness.

We prove AI trading can be both profitable *and* ethical — the agent doesn't just trade, it refuses.